Exercises - II

(1) Prove that the inverse and transpose of a symplectic matrix are symplectic.

(2) Find the Hamiltonian flow on $\mathbb{R}^2$ for the Hamiltonian function $H(x, y) = x + y$ and give a rough sketch of some integral curves in each case. Also, find the stationary points of the flows.

(3) Show that the integral curves of the Hamiltonian H on a symplectic manifold (M, ω) are contained in the level surfaces of H.

(4) Let ω_0 and ω_1 be two symplectic forms on a manifold M which are compatible with a complex structure J on M. Prove that ω_0 and ω_1 are homotopic. that is, there exists a path ω_t in the space of symplectic forms joining ω_0 and ω_1.

(5) Let $\pi : T^*M \to M$ denote the cotangent bundle and let λ be the tautological 1-form on M. Denote the symplectic form on T^*M by $\omega = d\lambda$.

(a) Show that a diffeomorphism $\varphi : M \to M$ naturally lifts to a symplectomorphism $\tilde{\varphi} : T^*M \to T^*M$ such that $\pi\tilde{\varphi} = \varphi\pi$, where .

(b) Let X be a vector field on a manifold M. Use the above observation to obtain a vector field $X_\#$ on T^*M whose flow on T^*M lifts the flow of X.

(c) Show that $X_\#$ is a hamiltonian vector field with hamiltonian function $i_{X_\#}\lambda$, where λ is the tautological 1-form on T^*M.

(6) If (M, ω) is a symplectic manifold then symplectic volume of it is defined by $\int_M \omega^n$ where M is oriented by ω^n. Show that symplectic volume is a symplectic invariant.

(7) *Let M be a closed manifold which is also oriented. If $\dim M = n$ then $H^n_{deRham}(M) \cong \mathbb{R}$ by deRham's theorem and the isomorphism is given by $\phi([\alpha]) = \int_M \alpha$. In particular, this implies that a top form α is exact if and only if $\int_M \alpha = 0$.*

Suppose that M and M' are two closed oriented manifolds of the same dimension with volume forms Ω and Ω'. Prove that the volume of (M, ω) is equal to the volume of (M', Ω'), (that is $\int_M \Omega = \int_{M'} \Omega'$) if and only if there exists a diffeomorphism $f : M' \to M$ such that $f^*\Omega = \Omega'$.